Blind Scalar Sector Selection and Reduced Source Construction

Metric-to-Action Mapping, Fisher–Energy Splitting, and a Reduced Scalar-Cosmology Demonstration

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July 28, 2026

Abstract

This paper gives a finite-dimensional source-side automation pipeline in which the scalar action and the resolved/unresolved sector split are constructed from a supplied background metric and an observation map. A diagonal metric model and optional reduction density determine the scalar action weights through the usual metric volume and inverse-metric factors. An observational Fisher form and a metric-dependent action Hessian then define the resolved sector by a generalized eigenproblem. The unresolved sector is eliminated at fixed resolved coordinates, giving an effective observation action W_{obs}. The construction is tested blindly on flat, random conformal, FRW, and Schwarzschild exterior radial backgrounds using the same mapper after the metric arrays are supplied. A reduced scalar-cosmology demonstration then shows how a smaller resolved sector preserves block observations far better than random sectors of the same dimension while still satisfying stationarity, envelope variation, and quotient-invariance checks. The result is not a derivation of a gravitational field equation; it is an automation protocol for choosing scalar source sectors and constructing reduced source actions on supplied backgrounds.

Contents

1 Purpose

The source-side action pipeline in its minimal form requires a metric-dependent scalar action and a resolved/unresolved split,

$$S[\phi; g], \qquad \phi = (\phi_R, \phi_N).$$

For practical adoption this is still too manual. A user should not need to guess the sector split. This paper therefore defines the preceding automation step:

$$\boxed{g, \mathcal{O} \longrightarrow S[\phi; g] \longrightarrow (R, N) \longrightarrow W_{\text{obs}}[\phi_R, g; \mathcal{R}].}$$

Here g supplies the scalar action geometry and $\mathcal{O}$ supplies the finite observation map.

Non-claim 1.1 (No metric dynamics)**.** The metric is supplied. This paper does not evolve the metric, derive the Einstein tensor, impose gravitational constraints, or claim an automated solution of the Einstein equations.

Non-claim 1.2 (No universal sector theorem)**.** The generalized eigen-split below is an explicit admissible and tested sector-selection rule. This paper does not prove that it is the unique possible resolution split in all field theories.

2 Metric-to-action map

Consider a finite $1 + 1$-dimensional diagonal Euclideanized metric model with arrays

$$g_{tt}(p) > 0, qquad g_{xx}(p) > 0,$$

and optional reduction density $q(p) > 0$, such as a shell-area factor in spherical reduction. Define

$$\sqrt{g}(p) = \sqrt{g_{tt}(p) g_{xx}(p)}.$$

The scalar action weights are

$$w_t(p) = q(p)\sqrt{g}(p) g^{tt}(p), \qquad w_x(p) = q(p)\sqrt{g}(p) g^{xx}(p),$$

and the potential/volume weight is

$$V(p) = q(p)\sqrt{g}(p).$$

The numerical scalar action is

$$S[\phi; g] = \frac{1}{2} \sum_{\langle pq \rangle_t} w_t(p)(\phi_p - \phi_q)^2 + \frac{1}{2} \sum_{\langle pq \rangle_x} w_x(p)(\phi_p - \phi_q)^2 + \sum_p V(p) \left(\frac{1}{2} m^2 \phi_p^2 + \frac{1}{4} \lambda \phi_p^4 \right).$$

This map is blind in the following sense: after the arrays (g_{tt}, g_{xx}, q) are supplied, the action is generated by the same formula for every tested background.

Proposition 2.1 (Background-blind scalar action map)**.** *For any positive diagonal metric arrays and positive density q, the above rule defines a convex Euclidean scalar action whenever $m^2 > 0$ and $\lambda \geq 0$. The field Hessian at the origin is*

$$H_g = K_g + m^2 \operatorname{diag}(V),$$

where K_g is the weighted graph Laplacian induced by w_t, w_x.

3 Observation geometry and blind sector split

Let

$$B : \phi \mapsto y$$

be a finite observation map. In the numerical tests B is a block-average sensor map. With unit noise covariance, the observational Fisher form is

$$F = B^T B.$$

The resolved sector is chosen by the generalized eigenproblem

$$\boxed{Fv_i = \lambda_i H_g v_i.}$$

For a budget k, define

$$R_k = \operatorname{span}\{v_1, \ldots, v_k\},$$

where eigenvalues are ordered descending. The unresolved sector is the H_g-orthogonal complement

$$N_k = R_k^{\perp_{H_g}}.$$

Definition 3.1 (Blind scalar sector mapper)**.** The blind sector mapper is the map

$$(g, B, k) \mapsto (S[\phi; g], R_k, N_k),$$

where $S[\phi; g]$ is generated by metric weights and (R_k, N_k) is generated by the Fisher–energy generalized eigenproblem.

Theorem 3.2 (Optimal Fisher capture under energy normalization)**.** *Let H_g be positive definite and F positive semidefinite. Among k-dimensional subspaces with an H_g-orthonormal basis, the span of the top generalized eigenvectors maximizes*

$$\operatorname{tr}(Q^T F Q), \qquad Q^T H_g Q = I_k.$$

Proof. Let $H_g = LL^T$ and set $u = L^T v$. The generalized problem is equivalent to the ordinary symmetric eigenproblem

$$L^{-1} F L^{-T} u = \lambda u.$$

For an H_g-orthonormal basis Q, the transformed basis $U = L^T Q$ is Euclidean orthonormal. The objective becomes

$$\operatorname{tr}(Q^T F Q) = \operatorname{tr}(U^T (L^{-1} F L^{-T}) U).$$

The Rayleigh–Ritz theorem states that this trace is maximized by the top k eigenvectors of the symmetric matrix $L^{-1} F L^{-T}$. Transforming back gives the top generalized eigenvectors of $Fv = \lambda H_g v$. □

Corollary 3.3 (Resolved sector as observation-visible energetic fiber)**.** *The resolved sector R_k is the maximally Fisher-visible k-dimensional sector measured in the action energy geometry H_g. The unresolved sector N_k is not a coordinate remainder; it is the energy-orthogonal complement of the selected observation-visible fiber.*

4 Effective observation action

After sector selection, write

$$\phi = Q_R a + Q_N b,$$

where Q_R and Q_N are H_g-orthonormal resolved and unresolved bases. For fixed resolved coordinate a, define

$$\boxed{W_{\mathrm{obs}}(a; g, B, k) = \min_b S[Q_R a + Q_N b; g].}$$

The hidden stationarity condition is

$$Q_N^T \nabla_\phi S[Q_R a + Q_N b^\star; g] = 0.$$

The metric/background response of the reduced model is obtained by varying W_{obs}. The envelope identity follows as in ordinary constrained minimization: derivatives of $b^\star$ do not contribute to first order because the unresolved gradient vanishes.

5 Blind background battery

The same mapper was tested on four supplied backgrounds: flat, random conformal, FRW conformal, and Schwarzschild exterior radial/tortoise. The background label is used only to generate arrays (g_{tt}, g_{xx}, q); the metric-to-action map and sector mapper are common.

Background	grid	metric error	stationarity rel.	envelope rel.	quotient error	auto/random capture
Flat	10^2	0	4.12×10^{-12}	7.64×10^{-10}	1.78×10^{-15}	1.000 / 0.249
Random conformal	10^2	0	1.57×10^{-9}	1.50×10^{-9}	0	1.000 / 0.249
FRW	10^2	0	1.03×10^{-11}	1.27×10^{-9}	1.78×10^{-15}	1.000 / 0.249
Schwarzschild exterior	10^2	0	6.32×10^{-13}	7.48×10^{-10}	0	1.000 / 0.252

Finding 5.1 (Blind sector construction)**.** The blind mapper constructs action weights and sectors without background-specific branching after the metric arrays are supplied. On all tested backgrounds, the automatically selected sector captures essentially all Fisher-visible directions at the chosen budget, while random same-dimensional sectors capture about one quarter. The subsequent unresolved-field elimination passes stationarity, local envelope, and quotient-invariance checks.

6 Reduced scalar cosmology demonstration

This section uses the blind mapper as an actual reduced model. The supplied background is the FRW conformal scalar model

$$ds^2 = a(\eta)^2(-d\eta^2 + dx^2),$$

with

$$a_{\mathrm{min}} = 0.7044, \qquad a_{\mathrm{max}} = 1.1857.$$

The grid is 12×12, so $n = 144$. The observation map consists of 2×2 block averages. A synthetic resolved physical field ϕ_0 is used only to provide observation values and resolved coordinates. The reduced action is then constructed by automatic sector selection and elimination of N_k.

k	Fisher capture	$W_{\mathrm{obs}}/S[\phi_0]$	obs. residual	field residual	envelope rel.	random obs. residual	random field residual
14	0.602	0.895	0.186	0.229	8.52×10^{-9}	0.909	0.922
25	0.837	0.951	0.014	0.126	2.75×10^{-10}	0.842	0.859
36	1.000	0.951	5.95×10^{-15}	0.126	3.16×10^{-10}	0.752	0.779

Finding 6.1 (First reduced scalar-cosmology use case)**.** At budget $k = 25$, the automatically selected sector captures about 84% of the Fisher-visible content and reduces the observation residual to 1.4%, compared with about 84% residual for random same-dimensional sectors. At $k = 36$, the selected sector captures the full block-observable content and the observation residual drops to numerical roundoff, while random sectors still miss most observations. In all cases the reduced action satisfies the envelope check.

Remark 6.2 (Why random sectors can have lower action)*.* Random sectors can produce lower minimized action because they may fail to preserve the observations or the physically selected resolved coordinates. The relevant adoption metric is therefore not only minimized action. It is simultaneous observation capture, stationarity, envelope consistency, and quotient invariance.

7 Protocol for use

A practical user can run the method as follows.

1. Supply metric arrays (g_{tt}, g_{xx}), optional reduction density q, and a scalar field family.
2. Generate action weights using $w_t = q\sqrt{g}g^{tt}$, $w_x = q\sqrt{g}g^{xx}$, and $V = q\sqrt{g}$.
3. Build the Hessian $H_g = K_g + m^2\,\mathrm{diag}(V)$.
4. Supply an observation map B and construct $F = B^TB$ or $F = B^T\Sigma^{-1}B$.
5. Solve $Fv = \lambda H_g v$.
6. Choose a budget k and define R_k from the top eigenvectors.
7. Eliminate N_k by minimizing $S[Q_Ra + Q_Nb; g]$ over b.
8. Use W_{obs} and its metric/background derivatives as the reduced source-side object.

8 Conclusion

This paper moves one step earlier than source-action elimination alone. The scalar action and sector split are now generated from a supplied metric and observation map. The sector split is not a hand-picked checkerboard or coordinate split; it is the top Fisher-visible subspace measured in the metric action geometry. The reduced scalar-cosmology example demonstrates a first usable workflow: automatic sector selection, unresolved elimination, reduced observation action, and quantitative comparison against random sectors.

A Reproducibility

The blind sector battery is implemented in `rg_blind_fiber_sector_automation_battery.py`. The reduced scalar-cosmology results are saved in `rg_reduced_scalar_cosmology_residual_results.txt`. The main numerical entry point is:

```
for bg in ['Flat','RandomConformal','FRW','Schwarzschild']:
    for N in [8,10]:
        print(run_case(bg,N,N))
```

The reduced cosmology diagnostic uses the same metric-to-action map and generalized eigen-sector selection, then compares automatic sectors to random sectors with the same dimension.

References

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[3] F. Zhang, *The Schur Complement and Its Applications*, Springer, 2005.